

1MCo5

Health Management Plan

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1 Purpose

The purpose of this Health Management Plan (HMP) is to describe how ALIGN will protect, maintain and improve health at work during Stage One Design Works and then for Stage Two. In addition, actions required to facilitate the development of the updated HMP and how health hazards will be evaluated and managed are detailed.

The HMP describes how ALIGN will meet HS2 requirements for occupational health through health hazard elimination and residual risk control using industry best practice at every opportunity.

ALIGN will demonstrate HS2's Safe at Heart Principles and implement measures to support HS2's health commitments:

To view health like safety, by:

- raising the profile of health with focus on harm prevention rather than mitigation as a principle;
- minimising the impact on our future workforce by focusing on health by design now; and
- working to a new improved standard for the provision of health care across the supply chain, leaving an industry legacy.

To eradicate wherever possible and where not possible to minimise exposure to the top five causes of long term occupational health disorders, which includes;

- focussing on the causes of occupational disorders relating to cancers, respiratory disorders, skin reactions, vibration and noise; and
- the selection of materials and handling methods to minimise workforce exposure to health hazards and risks.

2 Scope

This HMP sits within the ALIGN Health and Safety Management System and details how occupational health will be managed throughout the project at stage one and two. By using a systematic approach to review health issues, a health needs assessment will be used to utilise data gathered on the project. It will be undertaken through consultation and collaboration with the OHSP to define policy, process, people and responsibilities that will result in actions to realise the required outcomes. This document covers all ALIGN operating areas, projects and activities.

The overall prevention focussed strategy for occupational health is summarised within a 4W's programme as follows: -

Workplace

- Occupational health engagement with design teams to deliver the Health by Design requirements for the project.
- Profiling, tracking and reducing possible health hazards by the use of HHEMP and health risk assessment.
- Preventing occupational ill health through exposure to health hazards including:
 - Physical hazards such as noise and vibration;
 - Chemical & biological hazards including gas, vapour, fume, dust and bacteria;
 - Ergonomic stressors such as heavy and/or awkward loads and repetitive tasks;
 - Psychological factors such as pressure of work and working time
- Providing training to managers and employees, to ensure staff deal with health hazards in the same way as safety hazards and understand the importance of the hierarchy of control in reducing the risk of ill health.

Worker

- Where individuals have an existing health condition or where they experience ill health, we will provide support to help manage their condition and to remain in work, or return to work while supporting ongoing recovery
- Monitoring the health of individual workers to highlight:
 - suitability of controls (the need for health surveillance);
 - pre-placement fitness (pre-employment screening, fit for role medicals);
 - individual susceptibility (lifestyle and functional capacity assessments);
 - treatment of harm caused (first aid, and emergency treatment services)

Wellbeing

- Providing a range of activities and information in the workplace to promote healthy lifestyles and encourage employees to make and sustain positive behaviour changes. This includes implementing a yearly wellbeing calendar with monthly topics aligned with national health observances and seasonal wellbeing themes.
- Promoting well-being by using the workplace as a venue to promote healthy lifestyles.

Wider community

- Assisting wellbeing in the wider community by delivering initiatives that positively impact those surrounding our works. (such as health initiatives in local schools)
- Using the workplace as a platform to promote healthy lifestyles to the wider community, including employees' families, by sharing monthly wellbeing topics and

offering relevant resources and activities that extend the benefits of health initiatives beyond the workplace.

- Using information gained from national and local health data and local PCTs to ensure that initiatives and advice offered to both staff and the wider community is based upon relevant local health needs.

3 Definitions and Abbreviations

#	Abbreviations	Definitions
[1]	OHSP	Occupational Health Service Provider
[2]	HHEMP	Health Hazard Evaluation and Mitigation Process
[3]	HLF	Health Leadership Forum
[4]	OHMM	Occupational Health Maturity Matrix
[5]	HIFR	Health Impact Frequency Rate
[6]	CMP	Control and Monitoring Plan
[7]	SEQOHS	Safe Effective Quality Occupational Health Service
[8]	HRA	Health Risk Assessment
[9]	HSPI	Health and Safety Performance Indicator
[10]	UKAS	United Kingdom Accreditation Service
[11]	MERA	Medical Emergency Response Arrangements
[12]	OHI	Occupational Health Inspection

Responsibility for the creation and implementation of this health management plan will be managed, audited and assured by the ALIGN health, safety and wellbeing manager. Occupational Health management requires the support of a multidisciplinary team of occupational health professionals such as; occupational health advisors, occupational hygienists, ergonomists and other experts who will be deployed according to need to deliver this plan.

Occupational Health Lead

The Occupational Health lead is an integrated member of the Health and Safety team, supporting strategy development and implementation. Specific responsibilities include but are not limited to:

- Review health information and statistics from design reviews, fitness for work outcomes, health surveillance grouped data, Health Impact Frequency Rates, wellbeing outcomes etc. to assist in the identification of health needs and health assessment activity.
- With the assistance of the wider occupational health team, create and update HMP and HRAs using all available health data such as; information on worker health outcomes and demographics of the workforce and community.
- Creation of a monthly report on occupational health outcomes and information for Health and Safety Performance Indicators (HSPIs).
- Participation in the Health Leadership Forum (HLF).
- Manage service provision and ensure that outcomes are delivered in line with agreements and standards.

Occupational Health Service Provider (OHSP)

ALIGN has procured the services of Park Health and Safety Partnership (Park) to provide a SEQOHS accredited occupational health, safety, hygiene and wellbeing service designed to deliver HS2 requirements. The OHSP is responsible for;

- Supporting designers and the supply chain in the delivery of the requirements contained within this plan.
- Providing expertise to the team developing and running the Health by Design aspect of the programme, through reviews of documentation and training.
- Supporting the HHEMP process and providing advice on workplace interventions to control exposures that can lead to both acute and chronic occupational disease.
- Providing occupational health nurse led worker focused activities including fitness for work screening and medical assessment, health surveillance, clinical treatment services, emergency medical assistance, return to work assessments; rehabilitation and case management.

- Providing access to an Occupational Health Physician.
- Creating and running the wellbeing programme based on the health needs of the workforce.
- Retaining appropriate medical records, in accordance with legislation.

ALIGN's OHSP is accredited by Safe Effective Quality Occupational Health Service (SEQOHS) and has a management system that has been certified to ISO27001, ISO9001, OHSAS18001 awarded by a UKAS approved awarding body.

Design teams

Design teams shall ensure that they have access to appropriate ill health prevention advice so that occupational health considerations are specified within the design risk register (DRR). Designers shall ensure that any occupational health risks to construction, maintenance and operations staff are controlled in accordance with the principals of prevention and hierarchy of risk control.

Leaders

Leaders are responsible for ensuring the delivery of this Health Management Plan. This will be achieved by developing an effective occupational health service with the assistance of the OHSP. Leaders will ensure that appropriate occupational health resources are available to all projects, teams and workers in accordance with the requirements of this standard. Leaders shall ensure that resource for appropriate OH advice is available to all in the supply chain by reviewing the health needs for each package through use of the DRR and HRA and with advice from the OHSP.

Health Safety and Wellbeing teams

ALIGN shall ensure that all occupational health hazards have been identified (with the assistance of the OHSP) and the risk posed has been appropriately evaluated and controlled. Occupational health risk assessments and risk registers will outline how these can be eliminated or mitigated in accordance with the hierarchy of control to prevent occupational ill health occurring in the short, medium and long term.

Outputs from OH risk assessments and risk registers shall include any requirements for exposure monitoring and health surveillance dependant on risk level and in line with the client's specifications for reducing occupational exposures.

Suppliers

The supply chain shall ensure that they comply with, and that their employees participate in, all occupational health programmes in this health management plan. The Health and Safety and Wellbeing teams, in conjunction with the OHSP, will ensure the engagement of the supply chain through use of the health risk assessment process (see 5.2.2). Step 3 of this process requires specific responses from the supply chain regarding recognition of health risks, use of controls, exposure monitoring, and health surveillance requirements. The use of the Biosite system will record participation in training and briefings, as well as holding basic competencies and health records. This system will provide routine and exception reporting as to supply chain

engagement. The Orchid database used by the OHSP will provide further assurance of supply chain participation and will be used alongside Biosite to manage recall of operatives for both training and clinical assessment.

The Workforce

The workforce shall undertake work in compliance with approved systems that eliminate or reduce their risks to health. The workforce shall advise their line manager of any issues related to their fitness that might impact on them being able to perform their work activities. The workforce shall also advise their line manager of any circumstances that expose themselves or others to additional health risks and comply with work instructions to protect their, and others' health.

5 Health Management Plan

ALIGN will communicate the agreed standards to all contractors, who will be responsible for giving all workers access to the agreed standards. ALIGN will train managers to the agreed standards, managers will communicate the standards to all workers.

As the project staffing and activity profile develops the HMP will be reviewed and updated.

5.1 Health Monitoring

Where individuals have an existing health condition or where they experience ill- health, ALIGN will provide support to help manage their condition and to remain in work, or return to work while supporting ongoing recovery.

Monitoring the health of individual workers may include;

- Suitability of controls (the need for health surveillance)
- Pre-placement fitness (pre-employment screening, fit for role medicals)
- Individual susceptibilities (employees' lifestyle and functional capacity assessments)
- Pre-existing health conditions or susceptibility to ill-health arising from work
- Identification, assessment, referral, and advice provided regarding work and non-work related ill- health causes
- First aid, minor injury and treatments are logged. Trends will be identified, and cases followed up and monitored according to clinical guidelines
- Emergency response in the event of a significant incident in the workplace including assessment, referral and support provided to individual employees as required

Occupational health services will be provided that include, but are not limited to:

- Pre-employment screening
- Role identification and specific risk and health assessment related to job role
- Display Screen Equipment Assessment

- Return to work and rehabilitation services including information to access suitable health provision for subcontractors and other suppliers.

This system will ensure that workers can stay at work (by engaging with them through early identification of risk factors for health) and that where health issues are identified, through early engagement and rehabilitation (if required), workers can return to their role as soon as possible.

5.1.2 Fitness for Work Assessment

Fitness for Work assessments will be undertaken by the OHSP for all members of staff. Non-safety critical staff will be assessed through use of a questionnaire.

Fitness for Work Medicals will include pre-commencement health assessment (for all staff) and fit for role medicals for those undertaking **safety critical roles**. Supply chain will undertake their own fit for role medicals before commencing the project – unless with prior agreement

A safety critical worker is any person who, if incapacitated during the course of their task;

- has the potential to cause a significant safety risk to others and/or;
- due to the environment in which the task is undertaken, may put themselves at significant risk of harm.

This can include but is not limited to:

- Any mobile plant or machinery operators
- Hi-speed road workers
- Rail track-side workers
- Asbestos licensed workers
- Those working within a tunnel
- Those working within confined spaces
- Those working with live electricity
- Those working over open water
- Task carried out at height where collective preventative measures to control risk are not practicable, e.g.: harness work

All safety critical workers will be subject to additional job role specific medical assessment.

No person (including working visitors) is permitted to carry out a Safety Critical Role unless they have passed a Fit for role Medical (previously known as safety critical medical) carried out by their employer and can provide evidence of an appropriate existing fit for role medical valid for at least six months.

The minimum standard of a Fit for role Medical includes:

- A baseline health questionnaire to establish any current or previous medical / psychological health history.
- Blood pressure measurement
- Height, weight and body mass index
- Audiometry (hearing test)
- Spirometry (lung function test)
- Vision
- Musculoskeletal assessment

Fit for Role medicals will be valid for 3 years unless specified / detailed on the fit for role certificate or the job role requirements or individual's health determine otherwise.

If a person's fitness changes prior to expiry the individual has a responsibility to report this to their employer.

If there are any concerns with an individual's fitness for work arising from any health assessment a referral will be made to an appropriately qualified Health professional (e.g., GP, audiologist or own OH provider for surveillance). for further assessment.

ALIGN recognises that a number of industries employ medicals that cover the same requirements of a fit for role medical as identified above. ALIGN will undertake a review of these medicals on a case by case basis prior to making a judgement on their suitability for the project's needs.

5.1.3 Health Surveillance

Health surveillance is a system of ongoing health checks. These health checks may be required by law for employees who are exposed to noise or vibration, solvents, fumes, dusts, biological agents and other substances hazardous to health, or work in compressed air.

Health surveillance is not the same as health monitoring, health promotion or health screening.

- It should only be used for workers who need it:
- provides feedback about actions you may need to take to prevent further harm and protect workers
- allows workers to raise concerns about how work affects their health
- provides the opportunity to reinforce workers' training and education

Sub-contractors will be responsible for the management of health surveillance required for their own employees. There will be periodic audits carried out on the sub-contractors Occupational Health suppliers to ensure requirements are being met.

The Health Risk Assessment (HRA) will inform future needs for health surveillance.

The decision on the process for health surveillance will consider the following factors:

- Risk assessment and using effective controls are the primary control measure, negating the need for health surveillance where and whenever possible.
- Occupational health risk assessments to identify staff requiring periodic health surveillance will be undertaken including an assessment of ongoing health surveillance for operatives that may be provided by their employer.
- Health surveillance assurance audits will be carried out on the relevant subcontractors to ensure that their health surveillance strategies comply with legal requirements.
- ALIGN will ensure that operatives are available for required health surveillance assessment whether directly employed or as a sub-contractor.

Health risk	Type of health surveillance
Vibration	Tier 1 questionnaire (initial screening questionnaire)
	Tier 2 short questionnaire (annual screening questionnaire)
	Tier 3 – HAVS health assessment
	Tier 4- Formal diagnosis by a doctor in occupational health
	Tier 5- referral of the employee for certain tests for HAVS (is optional)
Noise	Audiometry (hearing checks)
	Examination by doctor if hearing damage identified
Respiratory	Respiratory surveillance questionnaire
	Lung function test
Respiratory (respirable crystalline silica)	Lung Xray
Skin(COSHH)	Assessing workers skin condition (baseline)
	Self-checking skin

	Then regular skin inspections by a 'responsible person'
Asbestos	Completion of medical surveillance form
	Completion of respiratory symptom questionnaire
	Clinical examination
	Chest Xray

5.1.4 Rehabilitation and Absence management

Vocational rehabilitation is whatever helps someone with a health problem to stay at, return to and remain in work. It is an idea and an approach, as much as an intervention or a service.

However, it can also be used too as an early intervention with conditions that are known to respond well to early support in the workplace such as depression, anxiety, stress, musculoskeletal disorders and long-term conditions.

Workers will be offered regular opportunities to interact with and get advice from the occupational health team in relation to their personal health to aid in early identification of health issues and be offered support appropriate to their needs.

Rehabilitation is a managed process that will be provided by the ALIGN OHSP or for the supply chain by their own OH provider if suitable services are available. The emphasis is not to force employees back to work but to take positive action in improving attendance and provide a better outcome for the employee, ALIGN will work with subcontractors to obtain sickness absence data to measure ill-health and absence trends.

ALIGN with the help of the OHSP will ensure that individuals with absences will be provided with an assessment to ensure they are fit to return to their role. Any adjustments to their role so that return to work is managed correctly will be outlined, including the provision for review of health assessments when required.

5.1.5 Records

ALIGN will ensure that where health records are required:

- These are completed within an appropriate timescale, in an approved and agreed manner.
- Contractors are informed, within an acceptable timescale, of the outcome of all health surveillance or fitness for role medicals.
- They are stored in a secure system for the appropriate statutory period, depending on the specific type of health record, ensuring all data protection requirements are met.

- All attendances by employees to Occupational Health will be fully documented in accordance with professional, legal and ethical requirements.
- Occupational health records will be legible, accurate and complete.

The process for recording information on health will acknowledge: -

- Medical Professional and Patient confidentiality
- The Data Protection Act 1998
- The Nursing and Midwifery Council Code of Practice
- Faculty of Occupational Medicine

Employee's occupational health information will be used to provide high-level statistical information. This information will always be anonymous, but the continuing collation and analysis of this data will be used to identify trends and make recommendations for improvement and development of the occupational health management plan and health activities.

5.2 Workplace

5.2.1 Health Hazard Evaluation and Mitigation Process

The following section describes the HHEMP for the project. The Health Hazard Evaluation and Mitigation Process is a structured approach to health hazard evaluation and mitigation ensuring that all individual health risks which cannot be eliminated are cascaded, mitigated and controlled.

- Hazard Identification: A programme of workshops and engagement sessions to identify hazards at all phases of the project.
- Evaluation of the risk these hazards present. These are also ranked to identify which risks should be prioritised for elimination or mitigation.
- Development of a system of recording this information and ongoing management of these documents.
- Adopting a clear strategy for addressing significant health risks for the project, taking opportunities to eliminate wherever possible and mitigate these where they cannot be eliminated.
- Ownership and communication of health risks throughout the organisations, transferring residual risks between design teams and construction teams and ultimately to the operator.
- Development and management of a process to manage health risks at a site level during the construction process.

Hazard Identification

The initiation of the HHEMP starts at the earliest opportunity from project conception and continues throughout the life of the project. This plan commences with a hazard identification exercise, both for project wide and on a site-specific level.

Hazard identification is carried out using several techniques, the primary of which are the HAZID workshops. These workshops involve disciplines leads including but not restricted to the OHSP, design leads, construction leads and safety managers. A programme of workshops will be established to cover the various activities of the project across the number of phases of the project. To ensure this is the case, HAZID workshops will be designed to cover: -

- Design, including constructability and logistic assessments
- Delivery, including construction activities and site setup
- Operations of the completed assets
- Planned, routine and one-off activities
- Demolition and/or decommissioning of assets

HAZID workshops will also identify the individuals who will likely be exposed to an identified hazard throughout the various phases of the project. These will include primarily the workforce undertaking the various phases of the project (construction operatives, maintenance workers etc.) but also a consideration of public exposure both within the communities being operated and those making use of the completed assets.

Design Input

Ongoing hazard identification will be continued throughout the design process with continual engagements between the health by design team, OHSP and the design teams. An integral element of this engagement is OHSP involvement in design review meetings to assist in developing an understanding of the current direction of the project and the demands of the design. This will allow the health by design team to provide meaningful feedback within: -

- Design Risk Register reviews
- Project Risk Register reviews
- Option Selection and Comparative Risk Assessment works
- Constructability Assessments

Whilst design engagement with the team will focus within current design meetings and reviews, specific workshops may also be carried out to drive on significant occupational health issues when identified.

Risk Identification

As with all risk assessment systems, occupational health hazards identified during the above processes will be assigned a risk score based on likelihood and severity. This will be recorded in

design risk registers and used in the compilation of HRAs. The scoring will be undertaken by the health by design team, and where required specialist expertise will be requested to advise on risk ratings. Based on both the hazard and risk scorings, entries into the HRA will be given a ranking to indicate the level of priority that the ALIGN project should apply to it. This ranking step is the final step of the evaluation process and will be undertaken by consensus of the team. The final ranking will identify a RAG score to each entry as follows: -

- **Green (Low)** – The level of risk is manageable with no additional controls during the relevant phase of the project. No further action required
- **Amber (Medium)** – The level of risk requires additional controls during the relevant phase of the project. Further action is required to reduce level of risk
- **Red (High)** – The level of risk requires multiple layers of control, including engineering and personal controls, to manage exposures. Action must be taken to reduce the risk

Elimination and Mitigation

The principle of elimination and mitigation will be undertaken following the Hierarchy of Control and Schedule 1 – General Principles of Prevention in The Management of Health and Safety at Work Regulations 1999.

Where a high ranked hazard has been identified which requires specialist knowledge of the nature of hazard and options for control, the health by design team will be used to call upon the expertise as and when required. All high ranked hazards will be subject to routine review by the team to ensure that the elimination and mitigation process is underway.

Option Review Workshops

Where a potentially difficult hazard has been identified, Option Review Workshops will be undertaken to discuss the various options available for mitigating this hazard. Such hazards that may be considered for this will be those where the mitigation or elimination of this hazard will introduce different health and safety risks. The action owner will lead this workshop, with assistance from the health by design team, to investigate the pros and cons of each option from and health and safety perspective. All parties who may have influence in this particular process should be present at this workshop. This workshop may identify a clear and obvious direction to be taken to mitigate the hazard, however it can be anticipated that there will be multiple alternative options, each of which present different types of risk. These should be evaluated by means of a Comparative Risk Assessment.

Comparative Risk Assessment

Comparative Risk Assessment is a means by which various materials, methodologies, and processes can be identified side by side using the same risk scoring system. It may be carried out as part of the option review workshops or independently where alternative options are easily identified. This should be an interdisciplinary exercise drawing on the expertise of all departments involved. Teams that may be required to assist with this may include; the health by design team, (including the use of specialisms such as; Human Factors and Ergonomics, Occupational Hygiene, Psychology, Community Impact, Clinical Occupational Health), health safety and environment, Procurement, Design, Construction and programme teams.

Part of the comparative risk assessment process will be a consideration of costings alongside the usual hazard and risks. It is therefore important that procurement is involved in the discussions when evaluating the viable options for mitigation. The output of a comparative risk assessment will give an indication of the option that should be preferred during the mitigation process. It should be noted that the comparative risk assessment will only be used as a guide for the mitigation process, and consideration of the effective management of residual risks may direct the hazard owner to opt for a higher hazard option. This may occur where more effective (engineering) controls can be chosen to manage residual risks, when a lower risk option has residual risks which can only be controlled by less effective controls (such as PPE).

Cost Benefit

The cost benefit of the implementation of the HHEMP should be understood and evaluated throughout the process in compliance with the Works Information. The cost of an intervention is difficult to evaluate as the cost of a change must be weighed up against the immediate costs of the alternative and any potential residual costing in the form of lost time, claims, and medical care.

For the basis of evaluating the cost of a design decision the following factors will be considered apply:

- Time saved or added in programme
- Use of specialist contractors
- Hire/purchase of specialist equipment
- Additional workforce requirements
- Other additional costs

These costs will be weighed up against the direct costs to ALIGN, specifically the costs of Occupational Hygiene and Clinical Interventions

- Requirement for surveys (evaluation and routine)
- Additional training requirements
- Engineering controls
- Provision of PPE
- Maintenance of controls
- Provision of SCMs
- Provision of Health Surveillance
- Provision of biological monitoring
- Additional treatment services
- Maintenance of records
- Lost time due to attendance at clinic

Further costings that ALIGN will consider in the cost benefit analysis will be the costs to the project should the exposures resulting from the design decision fail to be controlled adequately and resulting in an impact to a worker's health. These may include:

- Potential litigious claims
- Insurance

- Legal defence
- Compensation to the worker
- Loss of earnings to the worker

ALIGN will investigate the practicality of applying a costing to the UK economy, specifically with regards to the burden that the project has placed on the health service due to any ill health. These may include, but not be limited to;

- Cost of NHS Interventions
- Rehabilitation
- Ongoing treatment of clinical complaints
- Provision of specialist equipment (such as hearing aids?)

ALIGN appreciates the variety of costs that may be considered by different contractors and will therefore work with other Civils contractors to develop a single and uniform approach to cost benefit analysis across the HS2 project.

The delivery of this element of the HHEMP has been worked on in collaboration between ALIGN, HS2, and the other MWCCS and its development of the Health Impact Cost (HIC) tool. Once this has been formalised and released by HS2 it shall be used by the project to evaluate the health cost associated with its design.

BIM

ALIGN design teams will investigate the potential for using BIM to act as the primary resource for designers to be aware of the hazards being identified during each design stage. As hazards are identified by the above processes, a library of these could be inserted into the projects BIM system. Work is starting with our BIM Manager to ensure the model can and does receive the right inputs to allow it to function as the primary resource for capturing hazards identified during the design phase.

Continuous Improvement

A formal review of the HHEMP and its effectiveness will be undertaken on an annual basis by the ALIGN health and safety team.

5.2.2 Health Risk Assessments

The final step of the HHEMP is the transference of hazards and risks identified in the DRR into a site level health risk assessment. ALIGN will employ the following Health Risk Assessment system to achieve this.

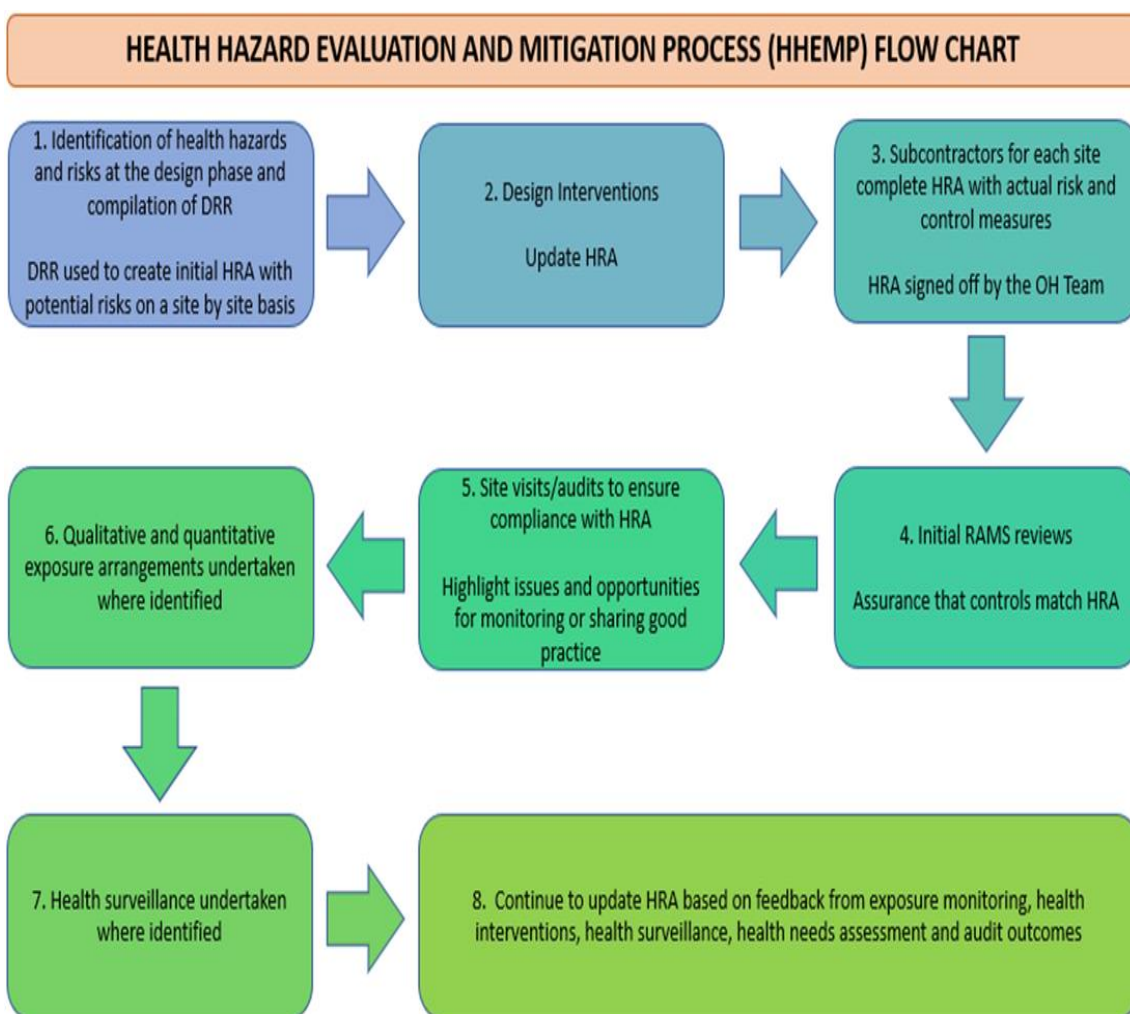
Each workplace and each work package will have a specific Health Risk Assessment (HRA) which will be updated regularly to consider the various elements of the project which may have an impact on the workers' health. In addition to identifying the health hazards of the work undertaken, and the controls required to reduce exposure, HRAs will also identify the requirements for workplace monitoring, health surveillance and training. Occupational Health Inspections (OHI) will be conducted at all ALIGN sites to assess whether the occupational health controls outlined in HRAs are being effectively implemented. The inspections will identify areas

where further monitoring may be required and provide insights into trends in compliance. A score will be generated based on the positive and negative observations from each inspection, offering a clear indication of how well the controls are being followed and highlighting areas for improvement.

The HRA will identify and group health hazards into distinct categories relevant to the project at the time. These will account for the Physical, Biological, Chemical and Psychological risks raised by the various elements of the works (such as existing conditions, ground contamination, biological risks, risks resulting from survey and construction activities and risks posed to and from the wider community).

The site teams planning and risk assessing works being undertaken on site should use the site HRA as the main resource for identifying health hazards and risks, as well as informing on controls and further occupational health requirements (such as monitoring and health surveillance). Where conditions on site lead to a differing level of risk (either increased or decreased) than that originally identified on the HRA, the health and safety team will be engaged to re-evaluate this risk and the HRA updated accordingly.

Where works will be undertaken by a subcontractor, the relevant HRA will be made available to them for consideration in writing their RAMS. Divergence from the measures identified in the HRA should be recorded in the relevant section, along with justification for the divergence.



5.2.3 Task Specific Risk Assessment

All activities undertaken must be subject to a specific health risk assessment to evaluate the potential exposures to the workforce and the means by which this is to be controlled. This will sit alongside the normal method statement and risk assessment. This must include, but not be limited to, the following elements:

- Identification / Grouping
- Evaluation of the likely level of hazard
- Duration of exposure and level of risk this poses
- Comparison of likely exposure against legislative limits
- Engineering control methods and efficiency of these
- Level of residual risk and PPE to control against this
- Requirements for exposure monitoring
- Requirements of health surveillance

Health risk assessments can be part of the overall risk assessment process, however health elements of these must be clearly marked as such and subject to the same requirements above. All health risks must be considered for elimination or substitution, evidence for which should be recorded in the Project Health Risk Register, before being incorporated into the task method statement and/or risk assessment.

Where a task specific risk assessment identifies additional health risks to a susceptible worker, the OHSP will work with ALIGN to ensure that the individual requirements of the worker are appropriately risk assessed and controlled. ALIGN will ensure that all those required to write risk assessments are provided training through the OHSP in the anticipation, recognition, evaluation, control and prevention of occupational health hazards. Further support will be provided through Occupational Health standards which outline the minimum requirements for a given task.

5.2.4 Workplace Stressors and Mental Health

Align recognise that psychological risks can often be overlooked in typical RAMS. In order to address this, we will introduce a holistic programme to improve and promote positive mental health.

In order to eliminate or reduce the impact of workplace stress and to comply with the legislative requirement for an organisational stress risk assessment, Align JV will apply the HSE management standards approach which, if managed properly, will reduce stress risk factors created within an organisation.

5.2.5

To proactively identify individual stress triggers and foster open communication between line managers and staff, Align will encourage the use of Wellness Action Plans (WAPs) across the project. This initiative will be incorporated into the Mental Health Training for Line Managers, which will be delivered throughout the Align project. Control of Exposure

Control of exposures to health hazards will be performed in line with;

- The Hierarchy of Control
- Schedule 1 – General Principles of Prevention – The Management of Health and Safety at Work Regulations 1999
- Schedule 2A – Principles of Good Practice – The Control of Substances Hazardous to Health Regulations 2002

Where exposure has been qualitatively assessed, or is based on a predicted level of exposure from previous projects, controls will be selected to ensure exposure does not exceed;

- limits identified in Appendix A of Wl1150
- limits below that listed in the EH40 or relevant legislation

ALIGN and its contractors must be cognisant of the exposure limits in Appendix A of Wl1150. These are rolling limits which become increasingly restrictive as the project progresses. These must therefore be regularly referred to as updates are made. It is particularly important that works planned in future stages of the project will require consideration of more restrictive exposure limits for noise, vibration and chemicals.

Control of exposure to chemicals will be considered as acceptable should the measured personal exposure be below the limits defined in Wl1150, or below a third of the legislative exposure limit - whichever is lower. Any exposure to a chemical in excess of half of the legislative limit will require PPE to control the residual risk.

High risk tasks with multiple health exposures, such as sprayed concrete lining operations, will be required to develop a specific Control and Monitoring Plan (CMP). This document will detail the specific controls required and how this will be managed throughout the process and project. The CMP will also outline the workplace monitoring requirements for the operation, including monitoring to evaluate exposures and ongoing reassurance monitoring.

5.2.6

Exposure Monitoring

Exposure monitoring is the method by which the level of exposure and the risk that this poses to the workforce is quantified. The OHSP will work with ALIGN to identify the exposures that require detailed quantification. This will primarily be based on the HRAs.

Monitoring will be undertaken by the OHSP. Monitoring will be primarily performed where the nature of the exposure and the risk that it poses is uncertain or unknown. This monitoring will

be used to identify the engineering and administrative systems that can be employed to reduce the exposure to acceptable levels.

Monitoring will also be undertaken to provide a level of assurance that the established controls are suitable for the task and working effectively. Should an operation be routine for the duration of the project then reassurance monitoring will be performed regularly throughout the project. The frequency of this assurance monitoring will be governed by the extent of the exposure and the residual risk to the workforce with the controls in place, and the risk posed by control failure.

All personal monitoring will be undertaken in line with the approved Method for Determining Hazardous Substances (MDHS). Results will be compared to the relevant limit listed in EH40, or Appendix A of W1150, whichever is the lower limit. Any required improvement to control will be in line with section 5.2.5 above.

5.3 Wellbeing Strategy

ALIGN is committed to actively promoting the wellbeing and health of all personnel employed on the project. The OHSP will play a major role in health promotion and will develop a programme to actively promote and deliver appropriate occupational health and wellbeing activities.

In order to achieve this, the ALIGN Management Team will adopt a pro-active stance with regard to worker health & wellbeing with the aim of empowering workers to adopt a responsible approach to measures within their own control to keep healthy and to create opportunities for assessment and education.

In order to empower and engage the workforce to enable a positive behavioural health change, ALIGN is committed to a wellbeing strategy which raises awareness of both work related and lifestyle health issues. The wellbeing strategy will be based upon a health needs assessment of the workforce and the communities in which the work will be delivered.

The wider context of workforce wellbeing including social, emotional, spiritual, environmental, occupational, intellectual and physical wellness will be acknowledged and incorporated into the wellbeing plan.

5.3.1 Wellbeing Plan - Campaigns and Activities

Campaigns and activities to raise awareness of both work-related and lifestyle health issues will be carried out on a regular basis. Some will focus on preventing/reducing the risk to workers acquiring Occupational Diseases (e.g., Respiratory disease e.g., Chronic Obstructive Pulmonary Disease (COPD), Leptospirosis, dermatitis, noise induced hearing loss, hand arm vibration, occupational injuries associated with manual handling and repetitive strain).

Other campaigns will be aimed at improving worker's general health, i.e., advice given on smoking, diet, alcohol, exercise, mental ill health, stress, and skin cancers etc.

The wellbeing strategy will also be aligned with identified national health priorities (such as obesity, ageing population etc.)

Regularly collected information on the demographics of both our workforce and the communities in which we will be operating will be used to inform the strategy along with trends identified from occupational health interventions with workers. In this way, we will ensure that all wellbeing activities undertaken are evaluated for their effectiveness and that learnings from these evaluations will be used to further develop and inform future programmes.

General wellbeing and health promotion activities will be made available to all personnel working on the project. Examples of these are as follows:

- Lifestyle screening will be available to all project personnel and will be aimed at the early detection and intervention of problems such as high blood pressure, high cholesterol and other cardiac risks.
- Drop-in or booked wellbeing clinics will be held which are available to all staff. This will be arranged via request and time in the occupational health providers' timetable and will be made available to all staff.
- Targeted Health Education - initiated following the identification of adverse trends from HRA, health surveillance or statistics from lifestyle screening.
- Behavioural functional fitness programmes - to address the ageing population and reduce musculoskeletal injuries and chronic pain.

Evaluation of the wellbeing programme will include the use of evidence-based data on:

- Engagement with the programme
- Short medium and long-term behaviour changes as a result of the programme
- Improved health status as a result of the programme

5.3.2 Wellbeing and Mental Health

During the project mental health and fatigue will be a focus for the wellbeing plan due to the nature of the work and the tasks being undertaken. To this end we will promote the training and use of mental health champions, incorporate resilience campaigns for managers and staff and targeted campaigns on fatigue and sleep hygiene, as well as conducting a fatigue survey across all sites on the Align project.

Sub-contractors and Align staff will have access to mental health first aider training throughout the project. As well as Mental health first aiders being available on all Align sites, ALIGN will also use the Mates in Mind campaign helping employees to identify poor mental health within themselves and others, and giving them the tools to seek further help.

5.3.3 Wellbeing Partnerships

The OHSP will work with ALIGN to engage with any local health services (such as GP's, walk-in centres, dental, chiropody, opticians etc.) to improve the workforce access to health services on and off site and to facilitate prompt referral, assessment and treatment. Information will be provided to the workforce to enable access to primary care services.

5.3.4 Wellbeing Trends

The OHSP will work with the project teams to understand any trends derived from medicals, clinics, accident and incident causes as well as other information that may inform the general health of those on the project. The OHSP will develop and deliver wellbeing campaigns targeting trends to positive effect.

The workforce will be asked to provide feedback on the efficacy and content of the wellbeing programme that will be acknowledged and evaluated. Behavioural health change data will be obtained following wellbeing interventions via questionnaire and interview.

5.4 Wider Community

Wherever possible general health information will be shared with the wider communities in which works are being undertaken (such as health initiatives in local schools). Advice and information on general health will be shared in a form which is accessible to worker's families and relevant for all communities.

6 Medical and Emergency Response Arrangements

Prior to start-up all work packages and workplaces shall undertake first aid risk assessments based on size and risk profile, enabling them to identify the needs of their workforce and to identify the need for treatment services.

There will be a project wide emergency response plan which all sites will adhere to which will identify potential medical emergency scenarios from risk assessments.

Emergency response provisions will include, (as appropriate):

- First Aiders trained in relation to site specific situations/hazards and updated on the basis of accident data review;
- First aiders trained in advanced life support (FREC) and involved in training with on-site emergency medical staff and off-site emergency services e.g., ambulance service, helicopter emergency services;
- Emergency staff responsible for liaison with offsite emergency services; and
- Regular liaison and planning with appropriate local offsite emergency services.

The emergency response plan will be subject to regular reviews following any significant changes in circumstances and at least annually.

In addition to this ALIGN will undertake mock scenario drills regularly, testing the logistical support needed (such as communication modes and back-up equipment and roles and responsibilities of incident and emergency response teams and transport), deploying resources during drills, undertaking scenario planning, and simulating events dealing with triage for multiple casualties.

Reporting arrangements for accidents and incidents (including those related to occupational health) are outlined in the ALIGN incident management plan.

ALIGN will comply with and facilitate HS2 assurance activities . ALIGN will keep records of formal inspections including deficiencies identified and corrective actions taken.

The ALIGN OHSP has comprehensive systems in place for clinical governance and audit. Elements of this system include performance and competency management of staff from a clinical and professional perspective, and supervision and training of staff including periodic clinical audit.

Assurance for occupational health performance will be ascertained for the project using the P4 system of subcontractor audits. This process is designed to investigate and interrogate an organisation and how embedded occupational health is in its culture and practices against the ALIGN HMP standards. This information will feed into the HS2 HSPIs. In addition to the P4 audits, Health Surveillance Assurance audits will be carried out on the relevant subcontractors to ensure compliance health surveillance requirements.

All those involved with working on the project will be provided Occupational Health training that reflects, and is relevant to, their job role. This training will centre on the occupational health risks of their work, and how to identify and prevent ill health as a result of exposure to those risks.

To aid delivery of this training, ALIGN will ensure relevant Occupational Health Training is included within the projects training matrix. This should be established and work in conjunction with the additional training requirements identified in HRAs.

In addition, all mandatory training will be reviewed with the assistance of the OHSP to ensure that where relevant information on health risks and controls are adequately covered.

Information, instruction and training will be provided using a variety of formats including face to face, e-learning, pictorial guidance etc. depending on the level of risk and the needs of the worker (for instance language, level of education, responsibilities) and utilising methods such as role-playing, live demonstrations, and round-table group discussions.

Training will be evaluated for effectiveness.

Appendix A of WI 1150 – Occupational Exposure

Hazard	Employer's standard for occupational exposure – from year 3	Employer's standard for occupational exposure – from year 5
Hand Arm Vibration (HAV) *1 (Control of Vibration at work regulations 2005)	<i>Employer's standard requires:</i> - No daily exposure to exceed 4.0 m/s² A(8) - All exposures above 2.0 m/s² A(8) daily exposure to be recorded, investigated, and managed - Exposure averaged over 1 week must be < 3.5 m/s² A(8)week	<i>Employer's standard requires:</i> - No exposure above 3.0m/s² A(8) - No exposure above 1.0m/s² A(8) - Exposure averaged over 1 week must be < 2.5 m/s² A(8)week
Whole Body Vibration (WBV) *2 (Control of Vibration at work regulations 2005	<i>Employer's standard requires:</i> - No daily exposure to exceed 0.8.00 m/s² A(8) - Exposure to frequencies in the 4-12Hz to be minimised - All daily average exposures above 0.3 m/s² A (8) to be recorded, investigated and managed. - Exposure averaged over 1 week must be < 0.80 m/s² A(8)week	<i>Employer's standard requires:</i> - No daily exposure to exceed 0.7 m/s² - All daily average exposures 0.2m/s² to be recorded, investigated and managed. - Exposure averaged over 1 week must be < 0.7 m/s²
Noise *3 (Control of Vibration at work regulations 2005	<i>Employer's standard requires:</i> Exposure limit values – No exposure above 86 dB(A) (daily or weekly average) or above 137 dB (C-weighted) without appropriate hearing protection.	<i>Employer's standard requires:</i> Exposure limit values – No exposure above 85dB(A) (daily or weekly average) or above 135 dB (C-weighted) without appropriate hearing protection.
Inhalable Dust *4	<i>Employer's standard requires:</i> No exposures above 7mg.m⁻³ (8hr TWA) without use of appropriate PPE.	<i>Employer's standard requires:</i> No exposures above 6mg.m⁻³ (8hr TWA) without use of appropriate PPE.

Hazard	Employer's standard for occupational exposure – from year 3	Employer's standard for occupational exposure – from year 5
General dust with no specific WEL (Regulation 2 of COSHH Regulations)	Action level of 4mg.m⁻³.	Action level of 3mg.m⁻³.
Respirable Dust *5 General respirable dust with no specific WEL (Regulation 2 of COSHH Regulations)	<i>Employer's standard requires:</i> - No exposures above 2mg.m⁻³ (8hr TWA) without use of appropriate PPE. - Action level of 1.5mg.m⁻³.	<i>Employer's standard requires:</i> - No exposures above 1mg.m⁻³ (8hr TWA) without use of appropriate PPE. - Action level of 1mg.m⁻³.
Respirable Crystalline Silica *6 (Regulation 2 of COSHH Regulations)	<i>Employer's standard requires:</i> - No exposures above 0.05mg.m⁻³ (8hr TWA) without use of appropriate PPE. - Action level 0.025mg.m⁻³.	<i>Employer's standard requires:</i> - No exposures above 0.025mg.m⁻³ (8hr TWA) without use of appropriate PPE (recognising that a more sophisticated sampling and analysis strategy may be required) - Action Level TBC
Lead *7 The Control of Lead at Work (CLAW) Regulations	<i>Employer's standard requires:</i> No exposures above 0.10mg/m⁻³ (8hr TWA), without use of appropriate PPE.	<i>Employer's standard requires:</i> No exposures above 0.05mg/m⁻³ (8hr TWA), without use of appropriate PPE.
Machine Made Mineral Fibres*8 (COSHH Regulations)	3mg.m⁻³ (8hr TWA) or 1fibres per ml (8hr TWA) Action Level: 1mg.m⁻³ (8hr TWA) or 0.5fibres per ml (8hr TWA)	1mg.m⁻³ (8hr TWA) or 0.5fibres per ml (8hr TWA)

Hazard	Employer's standard for occupational exposure – from year 3	Employer's standard for occupational exposure – from year 5
Refractory Ceramic Fibres *9 COSHH Regulations	0.5 fibres per ml (8hr TWA)	0.1 fibres per ml (8hr TWA)
DEEE *10 (Diesel Engine Exhaust Emissions) COSHH Regulations	- No exposures above 20ug/m³ EC, (8hrTWA) without use of appropriate PPE - Action level 10ug/m³. - Recommended use of RPE where exposure is >10ug/m³.	- No exposures above 10ug/m³ EC, (8hrTWA) without use of appropriate PPE. - Recommended use of RPE where exposure is >10ug/m³.
Bio-aerosols*11 COSHH Regulations	TBC	TBC
Wood Dust*12 COSHH Regulations	3mg.m⁻³ - Action level 2mg/m³.	2mg.m⁻³ - Action level 1mg/m³.

9 Reference Documents

#	Document Title	Reference
[1]	WI 1100 Health and safety	1MC01-HS2-PR-ITT-000-000471
[2]	WI 1150 Occupational health	1MC01-HS2-PR-ITT-000-000504